Handling Data 6 — Drawing and interpreting histograms

Calculator allowed; graph paper needed. Marks in brackets. ★ = harder question. Frequency density $= \dfrac{\text{frequency}}{\text{class width}}$; area of a bar = frequency.

1 The masses of 120 apples are recorded in the table. [10]

mass, m (g)	frequency
$60 < m \le 80$	12
$80 < m \le 90$	24
$90 < m \le 95$	20
$95 < m \le 100$	28
$100 < m \le 110$	26
$110 < m \le 130$	10

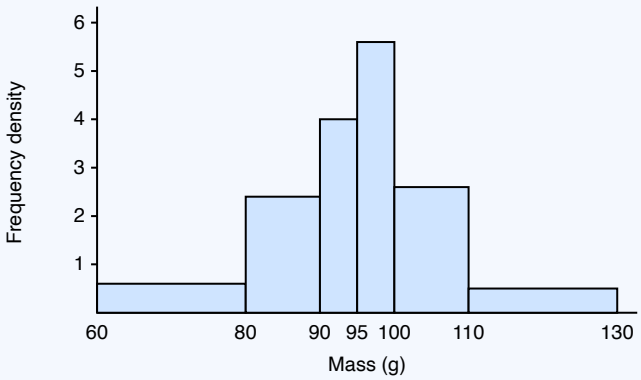
a Draw a histogram for this data.

b Estimate **i** the mean mass **ii** the median mass **iii** how many apples have a mass over 105 g.

c State the modal class.

a Work out each frequency density first (frequency ÷ width), then draw bars with *those* heights — unequal widths, no gaps.

class	frequency f	width	frequency density	mid-point x	fx
$60 < m \le 80$	12	20	0.6	70	840
$80 < m \le 90$	24	10	2.4	85	2040
$90 < m \le 95$	20	5	4	92.5	1850
$95 < m \le 100$	28	5	5.6	97.5	2730
$100 < m \le 110$	26	10	2.6	105	2730
$110 < m \le 130$	10	20	0.5	120	1200



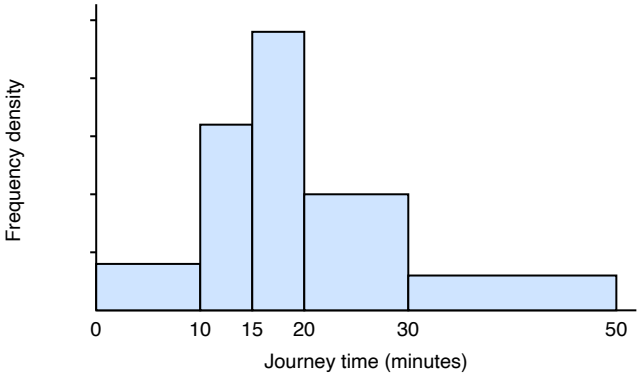
b i Mean $\approx \dfrac{\sum fx}{\sum f} = \dfrac{11390}{120} = 94.9$ g.

ii The median is the 60th value. Cumulative frequencies: 12, 36, 56, 84 — so it lies in the $95 < m \leq 100$ class, 4 values in. Estimate $95 + \frac{4}{28} \times 5 = 95.7$ g.

iii Over 105 g: half the $100 < m \leq 110$ class (13) plus all of the last class (10): about 23 apples.

c The modal class is the one with the highest frequency density: $95 < m \leq 100$ (not the class with the biggest frequency).

2 The histogram shows the journey times to school of the students in a year group. 8 students took 10 minutes or less. [15]



a Work out the frequency density of the first class and hence label the vertical axis.

b How many students are in the year group? **c** Estimate the median journey time.

d Draw a frequency table for the data. **e** Estimate the mean journey time.

f Estimate how many students take longer than the mean time.

a First class: width 10, frequency 8, so frequency density $= 8 \div 10 = 0.8$. The first bar reaches 0.8, so the grid lines are $1, 2, 3, 4, 5$ and the other bars are at 3.2, 4.8, 2.0 and 0.6.

b, d Frequency = density × width for each bar:

time, t (min)	frequency density	width	frequency	mid-point	fx
$0 < t \leq 10$	0.8	10	8	5	40
$10 < t \leq 15$	3.2	5	16	12.5	200
$15 < t \leq 20$	4.8	5	24	17.5	420
$20 < t \leq 30$	2	10	20	25	500
$30 < t \leq 50$	0.6	20	12	40	480

Total $= 80$ students.

c Median = the 40th value. Cumulative: 8, 24, 48 — in the $15 < t \leq 20$ class, 16 values in: $15 + \frac{16}{24} \times 5 = 18.3$ minutes.

e Mean $\approx \frac{\sum fx}{\sum f} = \frac{1640}{80} = 20.5$ minutes.

f Longer than 20.5 min: the part of the $20 < t \leq 30$ bar above 20.5 is $\frac{30 - 20.5}{10} \times 20 = 19$, plus the 12 in the last class: about 31 students.

Total: 25 marks

Check: histogram heights are frequency densities, not frequencies · frequency = density × width (area) · mean: mid-points × frequencies, divide by total · median: find the class of the $\frac{n}{2}$ th value, then go a fraction of the way through it · modal class = tallest bar · unequal widths → no bar may be drawn with its frequency as height.